

Plant-based whole-food diets are feasible during auto-HCT and are associated with dose-dependent microbiome modulation

Katherine Ueland,¹ Taara Elahi,¹ Mary Rasmussen,¹ Alex E. Wolfe,¹ Hayley Purcell,^{2,3} Saranya Rani Chakka,¹ Mercy Mirimo-Martinez,¹ Heather Persinger,¹ Karen Johnson,¹ Alanna Boynton,¹ Kerry McMillen,¹ Mariia Byelykh,³ Melinda A. Biernacki,^{1,3} Albert C. Yeh,^{1,3} Naveed Ali,^{1,3} Shivaprasad Manjappa,^{1,3} Natalie Wuliji,^{1,3} David N. Fredricks,^{1,3} Marie Bleakley,^{1,3,4} Leona A. Holmberg,^{1,3} Jonathan U. Peled,⁵ Jeannette M. Schenk,¹ Daniel Raftery,^{2,3} Jing Ma,¹ Geoffrey R. Hill,^{1,3} Marian L. Neuhaus,^{1,3} Stephanie J. Lee,^{1,3} and Kate A. Markey^{1,3}

¹Fred Hutchinson Cancer Center, Seattle, WA; ²Northwest Metabolomics Research Center, University of Washington, Seattle, WA; ³University of Washington, Seattle, WA;

⁴Seattle Children's Hospital, Seattle, WA; and ⁵Memorial Sloan Kettering Cancer Center, New York, NY

Key Points

- We have performed a clinical trial (n = 22 patients) of a plant-based, whole-food diet intervention to target the microbiome during auto-HCT.
- The intervention was feasible and was associated with dose-dependent modulation of the microbiome and an increase in fecal SCFA.

Plant-based whole food diets may represent a tractable approach to mitigating microbiome disruption and improving outcomes in patients undergoing autologous hematopoietic cell transplantation (auto-HCT) for multiple myeloma, a population in whom intestinal dysbiosis has been linked with inferior survival. We conducted a single-arm clinical trial at our center, in which participants undergoing auto-HCT (n = 22) received fresh, pre-prepared, plant-based meals for 5 weeks spanning conditioning, neutropenia, and early recovery, with the goal of supporting the consumption of nutrient-dense, high-fiber foods. The primary end points were feasibility and tolerability, defined by successful enrollment and patient-reported intake of study meals. Dietary intake was quantified using prospective food diaries and 24-hour dietary recall surveys. Secondary end points included changes in gut microbiome composition and function assessed by shotgun metagenomic sequencing and stool short-chain fatty acid (SCFA) measurements. The intervention was feasible and generally well tolerated, with all participants consuming delivered meals to some degree, with adherence sufficient to support planned dietary and correlative analyses. Greater intake of study meals was associated with more pronounced shifts in gut microbial communities, including enrichment of SCFA-producing taxa and compositional changes consistent with a fiber-responsive microbiome. Stool SCFA concentrations increased from baseline to the end of the intervention, suggesting a potential influence of the dietary strategy on microbial metabolite production during the peritransplant period. These findings demonstrate that a plant-based meal delivery intervention is implementable during auto-HCT and suggest dose-dependent modulation of the gut microbiome and its metabolic output. The trial was registered at [ClinicalTrials.gov](https://clinicaltrials.gov) as NCT06559709.

Submitted 2 March 2026; accepted 18 May 2026; prepublished online on *Blood Advances* First Edition 12 June 2026; final version published online 12 August 2026. <https://doi.org/10.1182/bloodadvances.2026020270>.

Metagenomic sequencing data from stool samples are available on the NCBI Sequence Read Archive database (<https://www.ncbi.nlm.nih.gov/sra>), under project ID: PRJNA1469158.

The full-text version of this article contains a data supplement.

© 2026 American Society of Hematology. Published by Elsevier Inc. Licensed under Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International (CC BY-NC-ND 4.0), permitting only noncommercial, nonderivative use with attribution. All other rights reserved.

Introduction

Patients undergoing autologous hematopoietic cell transplantation (auto-HCT) for multiple myeloma face significant risks, including impaired immune recovery, infections, and eventually, almost inevitable disease progression. Emerging evidence suggests that dietary intake can influence the gut microbiome, which, in turn, affects immune function.¹

The intestinal microbiome is markedly disrupted during cancer treatment and transplantation, and several clinical observational studies have demonstrated that the severity of this disruption may be a biomarker for outcome. In a large 2-center observational analysis (n = 534), we demonstrated that patients undergoing auto-HCT commence the transplantation process with fecal microbial diversity that is lower than that of healthy volunteers and that patients subsequently incur further microbiome disruption during the neutropenic period following transplantation.² Specifically, we observed that patients with greater microbiome damage, quantified by below-median fecal diversity during the perineutrophil-engraftment period, had higher rates of disease progression following transplantation and inferior overall survival compared with patients with preserved microbial diversity. Three other single-center studies reported similar findings.³⁻⁵

Previous work also suggests that plant-based eating patterns may benefit people across the myeloma spectrum. Large cohort analyses have found markedly lower incidence of multiple myeloma among pescatarians, vegetarians, and vegans compared with meat-eaters (70%-80% relative risk reduction, although based on few cases), suggesting a potential benefit from plant-rich diets even before diagnosis.⁶ A recent trial in patients with myeloma-precursor conditions tested a high-fiber, plant-based dietary intervention in overweight or obese individuals for 24 weeks.⁷ Provision of plant-based meals and lifestyle coaching was feasible and associated with improved quality of life (QoL), body mass index, and intestinal microbiome diversity. Complementary mechanistic experiments using the Vk*MYC mouse model of myeloma demonstrated that a high-fiber diet delayed disease progression via increased short-chain fatty acid (SCFA) production and enhancement of antitumor immunity, supporting the hypothesis that high-fiber, plant-based nutrition may be a viable strategy to slow disease progression in patients with myeloma.

Plant-based diets contain fermentable fiber, which is a critical substrate for immunomodulatory SCFA production by gut microbes. Plant-based diets that contain a large proportion of high-fiber foods such as fruits, vegetables, whole grains, legumes, nuts, and seeds typically have a higher nutrient density than western diets, and nutrient density has been demonstrated to lower the risk of cancer and all-cause mortality.⁸ Conversely, western dietary patterns, which tend to contain a large proportion of processed meats, high-fat dairy products, and refined grains, are thought to be proinflammatory and are associated with higher myeloma risk, reinforcing the rationale for emphasizing minimally processed plant-based foods.⁹

Despite strong observational links between microbial health and transplant outcomes, no microbiome-directed dietary interventions are currently integrated into routine auto-HCT care. The central hypothesis of this work is that increasing intake of diverse, fiber-rich, plant-based whole foods during the peri-auto-HCT

period will optimize microbial health, mitigate the most common transplant-related toxicities (including infections), and favorably influence immune recovery and longer-term outcomes. To inform the design of a definitive randomized trial, we conducted a pilot study in which we facilitated the delivery of plant-based whole-food meals to patients undergoing auto-HCT for myeloma, in order to encourage the intake of these high-fiber, nutrient-dense meals during the peri-HCT period. This pilot study demonstrated that the intervention was feasible and well tolerated by this patient population. Furthermore, we observed dose-dependent associations between consumption of study meals and changes in the microbiome, along with increased stool SCFA concentrations at the end of the intervention period compared with baseline, supporting further evaluation in larger, randomized studies.

Methods

Recruitment and enrollment

The trial protocol was approved by the Fred Hutchinson Cancer Center Institutional Review Board (FH/IRB0020501) and registered on [ClinicalTrials.gov](https://clinicaltrials.gov) (NCT06559709). Recruitment and screening for eligible participants occurred from October 2024 to July 2025. The recruitment flowchart is shown in supplemental Figure 1. Inclusion criteria are described in the study protocol (supplemental Methods). Outpatient auto-HCT is the usual practice at our center. All participants provided written informed consent for their participation in the study, including collection, storage, and research use of their biospecimens and associated clinical data, in accordance with the Declaration of Helsinki.

Additional control samples were obtained from participants enrolled in an institutional biobanking protocol approved by the Fred Hutchinson Cancer Center Institutional Review Board (FH/IRB000956). All participants provided written informed consent for the collection, storage, and research use of their biospecimens and associated clinical data. All authors had access to the primary clinical data. The analysis was performed by K.U. and K.A.M., and the statistical approach was developed in collaboration with and has the final approval of J.M.

Intervention

From day -7 to +28 (range, -11 to +30), participants received 3 meals per day plus snacks, delivered twice weekly by Thistle, a commercial meal-delivery service. All meals delivered to participants were completely plant-based (with no meat, eggs, or dairy) and included breakfast, lunch, dinner, and a daily snack. The menu changed weekly, and the study dietitian chose each participant's meals to adhere to the Fred Hutchinson Cancer Center food safety guidelines for immunosuppressed patients, which are largely based on the US Food and Drug Administration recommendations for safe food handling and preparation. Per the food safety guidelines, pre-packaged salads, pre-cut raw fruits and vegetables, and raw nuts and seeds were not permitted. Breakfast options included: tofu scramble, oatmeal, and high-protein muffins. Lunches were 1 of 4 rotating soups, and dinners consisted of a plant-based protein source (eg, tofu) with vegetables, legumes, and whole grains and pasta. Meals were delivered twice per week and required refrigeration and minimal preparation (eg, microwaving). Study participation did not mandate avoidance of meat, eggs, or dairy nor did it preclude nutrition support (eg, enteral or parenteral supplementation) if

clinically indicated. All study participants were regularly assessed by an oncology-trained registered dietitian, in accordance with usual care at our center. Nutritional guidance followed our usual clinical guidelines and was not altered for study participants.

Surveys, sample, and data collection

Dietary intake (of delivered meals as well as any other foods consumed) was monitored using a single 24-hour recall survey at baseline (before commencement of the food deliveries), nadir (day, 10 ± 5), and study end (day, 28 ± 5), and via self-reported multiday food records. Recall surveys were conducted by the Fred Hutchinson Cancer Center Nutrition Assessment Shared Resource and yielded 173 unique nutritional parameters. Patients were also asked to complete multiday food records at least 3 times per week throughout the study period, and these records were used to quantify consumption of the delivered meals.

Participants also met weekly with the study dietitian, and dietary intake and tolerability were qualitatively assessed during these visits. Stool samples were requested weekly for exploratory analyses (6 in total throughout the study period). Standardized collection kits were provided and patients returned the stool samples during clinic visits. Samples were processed within 24 hours and stored frozen at -80°C.

Assessments and end points

Primary end points were feasibility (enrollment, sample collection, and survey completion) and tolerability (meal consumption and qualitative patient feedback). We specified that we would deem the study feasible if we successfully recruited participants from our patient population and if ≥50% of the surveys and specimens were returned. To assess the tolerability of the intervention, the number of meals consumed, end-of-study (EOS) survey data, and qualitative feedback from the study participants were evaluated.

Secondary end points were all exploratory. Nutritional exploratory end points included the following: (1) adequacy of dietary intake, defined using accepted clinical nutrition guidelines (European Society for Clinical Nutrition and Metabolism)¹⁰ for calorie (30 kcal/kg per day) and protein (1.5 g/kg per day) recommended minimums for this patient group, and 24-hour dietary recall survey data were analyzed using the Nutrition Data System for Research software version 2024, developed by the Nutrition Coordinating Center, University of Minnesota, Minneapolis, Minnesota; and (2) diet quality was assessed using the Healthy Eating Index-2020 (HEI-2020), modeled as a function of time and the intake of delivered meals. The HEI-2020 is a validated measure of overall dietary quality, and scores were generated by the Nutrition Data System for Research software based on participants' 24-hour dietary recalls. Microbiome-related exploratory end points included microbiome diversity and composition (shotgun metagenomics) and SCFA concentrations in stool (metabolomics). Clinical data to support the analysis of exploratory end points were prospectively collected. Patient-reported outcomes were assessed using the European Organisation for Research and Treatment of Cancer (EORTC) Quality of Life Questionnaire-Myeloma Module (QLQ-MY20) and the Core Module (QLQ-C30) at baseline and day 28.^{11,12}

Please refer to supplemental Methods for metagenomic sequencing, metabolomics, computational analyses and statistics, software, and data sharing.

Results

The study intervention was feasible and well tolerated

Twenty-two patients were recruited over a 10-month period (Table 1; Figure 1A). More than 50% of all planned biospecimen samples and patient surveys were successfully collected, indicating that the study procedures were feasible as defined in the study protocol (individual patient timelines; Figure 1B). There were no withdrawals attributed to study-related procedures, and no adverse effects of the meal deliveries. Three-month restaging studies confirmed that patients with available data (n = 18/22) were in complete remission.

Of the 22 patients who commenced the study, 17 received all 5 weeks of intended food deliveries (median days on study, 35; range, 7-35). An average of 112 meals were delivered per participant (range, 88-135). Thirteen patients required hospital admission for symptom management and did not consume the delivered meals during their inpatient stay. These admissions occurred during the neutropenic nadir period, where regimen-related toxicities limited intake (anorexia [65%], diarrhea [88%], dysgeusia [71%], mucositis [59%], and nausea [82%]). Three patients (18%) received parenteral nutrition.

Table 1. Patient characteristics

Demographics of the study population	n (%)
Age (median, range), y	58.5 (42-72)
Sex	
Female	8 (36)
Male	14 (64)
Race	
Asian	1 (4.5)
Black or African American	3 (14)
White	17 (77)
Native Hawaiian or Pacific Islander	1 (4.5)
Ethnicity	
Hispanic or Latino	0 (0)
Non-Hispanic or Latino	22 (100)
R-ISS at diagnosis*	
Stage 1	4 (20)
Stage 2	13 (65)
Stage 3	3 (13)
Disease stage at transplant	
CR	1 (5)
PR	3 (13)
VGPR	18 (82)
ECOG performance status	
0	8 (36)
1	13 (59)
2	1 (5)

CR, complete remission; ECOG, Eastern Cooperative Oncology Group; PR, partial response; R-ISS, Revised International Staging System; VGPR, very good partial response. *Data missing for calculation of R-ISS stage for 2 participants, hence n = 20 was used to calculate values.

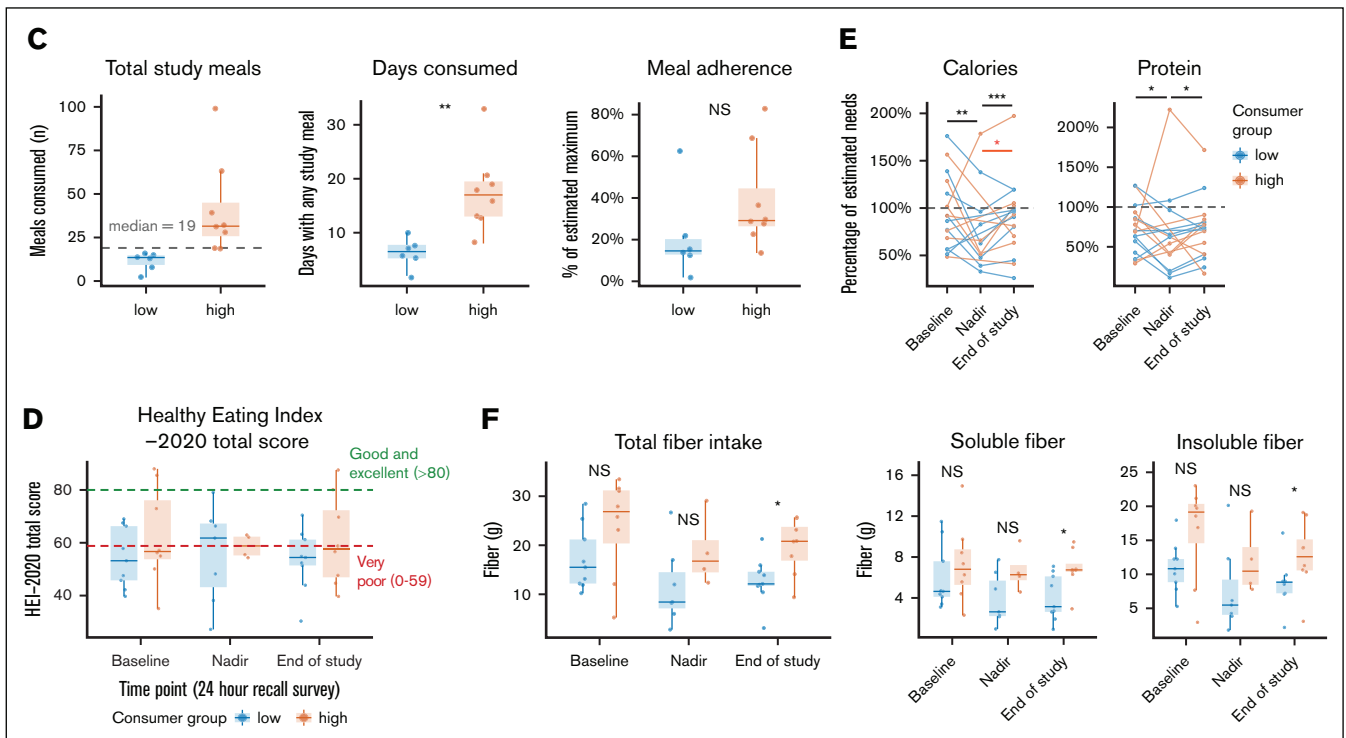


Figure 1 (continued) display the following: neutropenic period (orange bars); hospital admission (blue bars); antibiotic exposure (cefepime, green bars; piperacillin-tazobactam [Zosyn], teal bars); stool specimen collection (green diamonds); nutrition (24-hour dietary recall performed, green triangles; food record available, purple squares; meal delivery, pink inverted triangles; TPN, pink bars); and fiber intake (orange circles scaled by daily fiber consumption: <5 g, 5-10 g, 10-20 g, and >20 g). The display window spans day –14 to day +42 to focus on the peritransplant period. Twenty-nine data points from 10 included patients that fell before day –14 (range, –51 to –15; primarily pretransplant stool samples, dietary recalls, and fiber intake measurements) were omitted for visual clarity. (C) Summary of meal consumption by total number of meals, days consumed, and proportion of total theoretical maximum, for 14 patients of the 17 who remained on the study and who returned food records. Line shown at median in the left panel indicates total meals consumed. Low consumers are represented by blue dots and shading; high consumers are represented by orange dots and shading. (D) HEI index at baseline (BL), nadir, and EOS, calculated from 24-hour recall surveys. Dashed line highlights the cut-point for the “very poor” range (<60). (E) Proportion of recommended intake of calories and protein at BL, nadir (day 10), and at the EOS (day 28), consumed by each patient, colored by consumption group. Intake of these nutrients defined by 24-hour recall surveys. Dashed line highlights the 100% point. (F) Fiber intake status in high and low consumers over time. Statistical significance is indicated as * $P < .05$, ** $P < .01$. NS, not significant; TPN, total parenteral nutrition. Panel A was created in BioRender. Markey K. (2026) <https://BioRender.com/jvpx3ue>.

The median number of delivered meals that participants reported consuming was 19 (range, 2-99; [Figure 1C](#)). The variation in meal consumption among patients provided an opportunity to divide the study cohort into “high” and “low” meal consumers (above and below median number of meals consumed; [Figure 1C](#)) and explore the association between meal consumption and nutritional parameters in further analyses. Several strategies yielded very similar groupings, including absolute meals consumed, the number of days on which the plant-based meals were eaten, and the proportion of total meals consumed (of maximum delivered; [Figure 1C](#)). We elected to use the above- and below-median absolute number of meals consumed as the basis for designation as either a high or low consumer. For 3 patients who did not return food records, qualitative records from weekly interviews with the study dietitian allowed us to classify them into the low-consumption group. Despite variability in consumption across the different phases of auto-HCT, we deemed that the intervention was well tolerated overall.

Total caloric intake decreases between baseline and the neutropenic nadir period, and increases by EOS in association with high consumption of the plant-based study meals

We next used the 24-hour recall studies to assess dietary quality throughout the study period. We used the HEI-2020, which is a scoring system that measures how closely a person’s overall diet aligns with the Dietary Guidelines for Americans, providing an overall diet quality score from 0 to 100. The average score in the general population is in the “very poor” range (<59), and our patient group was similar to the general population, with scores largely in the “very poor” range that did not significantly change over the study period ([Figure 1D](#)).¹³⁻¹⁵ There was no improvement in the total HEI throughout the study period nor were there differences between the high- and low-consuming participants in any subcomponent score (supplemental Figure 2). Next, we assessed the overall calorie and protein intake of our study participants at baseline, during the

neutropenic nadir and at the EOS. As expected,¹⁶ calorie and protein intake decreased during the neutropenic nadir compared with baseline (Figure 1E). Patients reported consuming a primarily liquid diet of oral nutrition supplements and soups (nonstudy foods) during this time window. Interestingly, high consumers of the delivered meals had a higher calorie intake relative to predicted needs at the EOS compared with their neutropenic nadir, a pattern not seen in the low consumers (Figure 1E).

A key goal in delivering plant-based meals was to increase fiber intake, and we observed that high consumers of the plant-based meals indeed had a significantly higher total fiber intake (both soluble and insoluble fractions) at the EOS time point relative to the low consumers (Figure 1F). We next analyzed the 173 other nutrients captured by the 24-hour recall surveys. In this unbiased analysis, fiber was the predominant nutritional factor that differed between the high and low consumers of the plant-based meals (supplemental Figure 3).

α -Diversity is preserved throughout the study period

We next assessed fecal microbial α -diversity (using Shannon, Inverse Simpson, and Simpson indices) as well as observed richness at both strain and species resolution throughout the study period (Figure 2A). There were no differences between high and low meal consumers at any time point at either species or strain rank, suggesting a maintenance of α -diversity in this small cohort. Eight patients (47%) were exposed to cefepime during the peritransplant period (average duration 1.65 days [standard deviation, 2.18 days]), and only 1 patient received piperacillin-tazobactam (1 day duration). No patients received carbapenem antibiotics during the study period.

Community-level β -diversity does not differ between groups

To extend our analysis of the microbiome, we next visualized microbial composition in study participants and an additional 15 nonstudy comparator patients with pre-auto-HCT myeloma (stacked bar plots at family rank; Figure 2B). Baseline microbiome composition was similar among subsequent high and low consumers, and trial participants did not differ in microbiome community composition from nonstudy comparators (3-group permutational multivariate analysis of variance [PERMANOVA] comparing control participants with high- and low-consuming study participants was not significant at baseline; $R^2 = 5.8\%$; $P = .217$; Figure 2C).

Similarly, when we analyzed post-auto-HCT samples, PERMANOVA analysis confirmed that consumer status explained only 2.4% of total compositional variance ($R^2 = 0.024$; $P = .338$), day relative to transplant explained 0.8% ($R^2 = 0.008$; $P = .940$), and the group-by-time interaction was similarly nonsignificant ($R^2 = 0.022$; $P = .520$). The residual fraction accounted for 94.7% of variance, reflecting the dominant influence of interpatient variability. Together, these findings confirm that in this small study, we cannot define community-level compositional shifts associated with the degree of consumption of the plant-based meals.

Unbiased taxonomic analysis reveals a number of differentially abundant strains after auto-HCT

Next, we asked whether specific phylogenetic branches of the microbiome were differentially represented among high and low consumers. We first generated taxonomic heat trees displaying

$\log_2$ fold-changes between high and low consumers (on a per-patient basis, all relative to each individual baseline sample). These visualizations, in which branches and nodes represent taxa that are then sized by strength of association and colored by consumption group, revealed a pattern of compositional differences (Figure 2D). The most prominent observations in this analysis were apparent differences within the Bacteroidota phylum (formerly Bacteroidetes; enriched in low consumers) and among many members of the Lachnospiraceae family (enriched in high consumers).

To further this exploratory analysis, we ranked the taxa by largest fold-change from baseline during neutropenic nadir and at EOS using genus, species, and strain-level data. The top 16 taxa (ranked by unadjusted P value) are shown in Figure 2E and summarized in Table 2. Among taxa enriched relative to baseline in high consumers, the largest effects were in SCFA-producing Lachnospiraceae (*Lachnoclostridium B* sp900066555; an unclassified Lachnospiraceae strain GCA_900757145) during the neutropenic nadir window, and the fiber-fermenter *Bifidobacterium longum* at EOS. Bacteroidetes (*Phocaeicola*, *Bacteroides uniformis*, and *Bacteroidaceae* sp902362375) and the pathobiont *Streptococcus mutans* were enriched in the low consumers. Due to the large burden of multiple comparison testing (~7300 tests across genus, species, and strain taxonomic ranks), no findings in this hypothesis-generating analysis were robust to false-discovery rate (FDR) correction. These findings support the hypothesis that higher plant-based meal consumption during HCT is associated with preferential recovery of some fiber-fermenting and SCFA-producing strains, and suppression of Bacteroidota/Bacteroidetes and at least 1 pathobiont (*Streptococcus mutans*).

Plant-based meal intake is associated with preservation of commensal taxa and suppression of pathobionts

To explore which specific nutrients may be associated with microbial community structure changes during the peri-auto-HCT period, we performed 2 additional analyses using dietary records matched to stool samples (Figure 3A). For 49 of the stool samples collected from study participants, we had detailed dietary data available from records collected in the previous 72 hours (either from self-reported food diaries or formal 24-hour recall studies). These data included all foods consumed (study and nonstudy). First, we asked whether fiber intake was correlated with any microbial taxa (at genus, species, or strain rank) and second, we performed an unbiased screen of all recorded nutrients and all genera followed by species/strain resolution of significant hits (Figure 3A).

Fiber-microbiome correlation analysis highlights strains associated with fiber intake in the setting of microbiome damage during chemotherapy and recovery

We first assessed whether fiber intake as a continuous variable correlated with overall community structure. PERMANOVA on Bray-Curtis distances showed that fiber intake explained just 2.3% of strain-level microbiome variance ($R^2 = 0.023$; $F = 1.13$; $P = .275$). This is consistent with large microbiome cohort studies, where overall dietary patterns explain ~5% to 20% of variance,

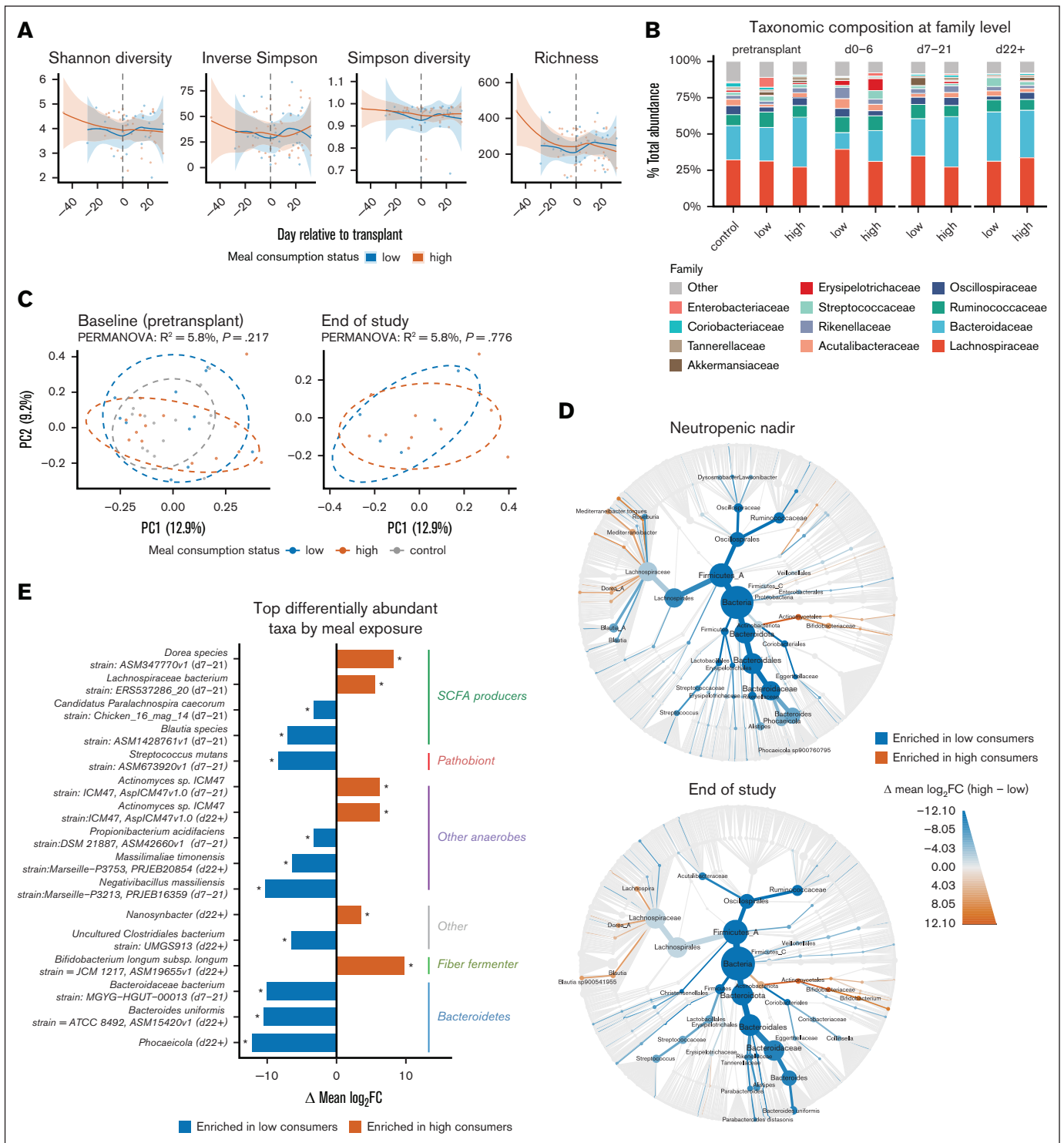


Figure 2. Plant-based meal consumption alters the intestinal microbiome during the post-auto-HCT recovery period. (A) α -diversity over time by diet exposure. Shannon diversity, Inverse Simpson, Simpson diversity, and observed species richness plotted (species level shown). Each dot represents 1 sample, curves are locally estimated scatterplot smoothed (LOESS) curves (span = 0.85) with 95% confidence bands. Blue indicates low meal consumers and orange indicates high meal consumers. Dashed vertical line indicates day of transplant (day 0). No significant differences were observed between high- and low-consumption groups at any time point, at either strain level (Shannon: all Wilcoxon $P \geq .59$; Simpson and Inverse Simpson: all $P \geq .53$; richness: all $P \geq .42$) or species level (Shannon: all $P \geq .59$; Simpson and Inverse Simpson: all $P \geq .59$; richness: all $P \geq .42$). (B) Taxonomic composition at family level by consumption group and time window. Bars show mean relative abundance. (C) Principal coordinates analysis of Bray-Curtis dissimilarities at species level at BL (control, high, and low consumers shown) and at EOS (high and low consumers). PERMANOVA on Bray-Curtis distances: no significant difference in BL composition between high, low, and control groups ($R^2 = 5.8\%$; $P = .217$) or when performing a 2-way comparison of high vs low

Table 2. Differentially abundant taxa in high and low consumers of the plant-based study meals

Taxon*	Grouping	Time window, d	Δ log ₂ FC	P value	Direction in high
<i>Phocaeicola</i> (genus)	Bacteroidetes	22+	−12.2	.035	Decreased
<i>Bacteroides uniformis</i> ATCC 8492	Bacteroidetes	22+	−10.5	.023	Decreased
<i>Negativibacillus massiliensis</i>	Other anaerobe	7-21	−10.3	.028	Decreased
<i>Bacteroidaceae</i> sp902362375	Bacteroidetes	7-21	−10.0	.037	Decreased
<i>Bifidobacterium longum</i> subsp. <i>longum</i> JCM 1217	Fiber fermenter	22+	+9.8	.035	Increased
<i>Streptococcus mutans</i>	Pathobiont	7-21	−8.4	.034	Decreased
<i>Lachnoclostridium</i> B sp900066555	SCFA producer	7-21	+8.2	.023	Increased
<i>Blautia</i> sp. NSJ-34	SCFA producer	7-21	−7.0	.021	Decreased
Uncultured <i>Clostridiales</i> sp900546685	Other anaerobe	22+	−6.6	.018	Decreased
<i>Massilimaliae timonensis</i>	Other anaerobe	22+	−6.4	.018	Decreased
<i>Actinomyces</i> sp. ICM47 (GCF_000278725)	Other anaerobe	7-21	+6.3	.035	Increased
<i>Actinomyces</i> sp. ICM47 (GCF_000278725)	Other anaerobe	22+	+6.3	.035	Increased
<i>Lachnospiraceae</i> sp900757145	SCFA producer	7-21	+5.6	.037	Increased
<i>Nanosynbacter</i> (genus)	Other	22+	+3.5	.030	Increased
<i>Candidatus Paralachnospira caecorum</i>	SCFA producer	7-21	+3.3	.036	Decreased
<i>Propionibacterium acidifaciens</i>	Other anaerobe	7-21	−3.2	.027	Decreased

FC, fold-change.
*Table sorted by absolute Δ log₂FC (largest effect first). Δ log₂FC = mean per-patient log₂FC in high consumers minus mean in low consumers; positive values indicate relative enrichment in high-consuming patients. All P values from Wilcoxon rank-sum test.

with individual dietary variables generally contributing only modest effect sizes in the order of a few percentage points each.^{17,18}

We next explored correlations between fiber intake and taxon relative (Figure 3A-B). Among the top positively associated taxa was a strain of *Blautia* A, a known SCFA producer (list of top taxa; supplemental Table 1). The most prominent negatively associated taxa included the oral origin pathobiont *Streptococcus constellatus* A, further supporting the hypothesis that fiber intake may support colonization resistance against opportunistic organisms.

Unbiased nutrient analysis identifies additional dietary components associated with microbial taxa abundance

To determine whether nutrients beyond fiber were associated with microbiome changes, we performed an unbiased 2-stage screening with 61 dietary variables against all genera (stage 1), followed by species/strain resolution of significant hits (stage 2).

Across the top 30 nutrient:taxa associations, Lachnospiraceae was the dominant family, contributing 18 of the 30 (60%) most significant pairs and spanning every nutrient category tested (Figure 3C; supplemental Table 2). We observed the strongest correlations between fiber-derived substrates and known butyrate-producing or saccharolytic clades, including *Anaerostipes hadrus* with both starch and omega-3 ($\rho = 0.49, 0.49$; P (unadjusted) = $5.8 \times 10^{-4}, 4.0 \times 10^{-4}$), *B. massiliensis* A with polyunsaturated and omega-6 fatty acids ($\rho = 0.50, 0.53$; $P = 2.8 \times 10^{-4}, 1.1 \times 10^{-4}$), and *Mediterraneibacter faecis* with omega-3 ($\rho = 0.48, P = 5.4 \times 10^{-4}$). Furthermore, within the Lachnospiraceae family alone, a total of 76 nutrient:taxon pairs reached an unadjusted P value of $<.01$, further suggesting that this family bears many potentially diet-responsive strains.

One association reached statistical significance after FDR correction across all tested taxa, a positive correlation between dietary alcohol and a strain of butyrate-producing *Roseburia*, which is a member of the Lachnospiraceae family

Figure 2 (continued) consumers ($R^2 = 5.1\%$; $P = .186$). Overall composition did not differ significantly by time period alone ($R^2 = 2.9\%$; $P = .931$), nor was there a significant group $\times$ time interaction ($R^2 = 4.07\%$; $P = .805$). N summary (unique patients); BL: control $n = 15$, low $n = 9$, high $n = 8$; day 0-6: low = 4, high = 5; day 7-21: low = 5, high = 6; day 22+ low = 5, high = 7. (D) Circular cladograms displaying the full strain-level taxonomy at neutropenic nadir and EOS. Node and edge color represent the aggregated log₂ fold-change in mean relative abundance between high and low consumption groups: orange indicates enrichment in high consumers, blue indicates enrichment in low consumers, and gray indicates no difference. Node size is proportional to mean total abundance. Data represent species rank and named/highlighted taxa met nominal significance value of $P < .1$ (61 taxa at nadir and 44 at EOS). (E) Diverging bar chart showing the difference in mean per-patient log₂ fold-change from BL (Δ mean log₂ FC; high–low consumer) for the 16 most significant taxa across 2 post-transplant time windows (day 7-21 and EOS). Bars extending to the right (orange) indicate taxa with greater enrichment (or less depletion) from BL in patients on a high-fiber diet and bars extending to the left (blue) indicate taxa with greater depletion in patients on a high-fiber diet. Colored labels on the right indicate microbial groupings (see supplemental Methods for definitions). Taxa are displayed at strain level with genome taxonomy database genome accessions, except *Nanosynbacter* and *Phocaeicola*, which were significant at genus level only. The 16 taxa shown were selected as the most significant after deduplication across taxonomic ranks, from a total of 7351 Wilcoxon rank-sum tests across 608 genera, 2188 species, and 2194 strains at 2 posttransplant time points (day 7-21 and EOS). After Benjamini-Hochberg correction across all taxa tested within each time point, no individual taxon reached significance. Unadjusted P values are as shown. Sample sizes per comparison: day 7-21, high $n = 6$, low $n = 5$; day 22+, high $n = 7$, low $n = 5$.

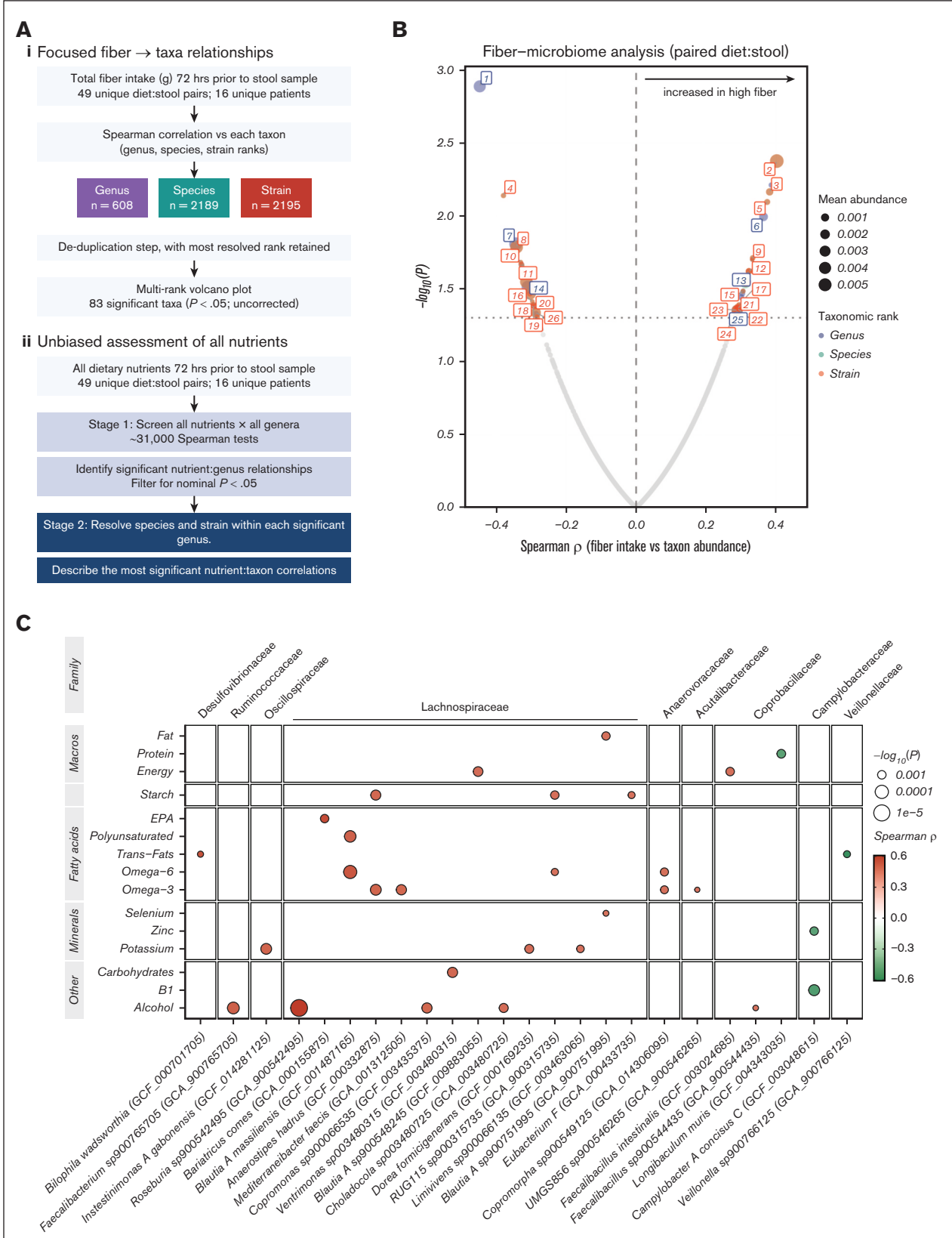


Figure 3. Specific nutrients correlate with bacterial strains during recovery from injury. (A) Schematic of analysis approach. (B) Volcano plot showing Spearman rank correlations between dietary fiber intake (g) and the relative abundance of each taxon at genus (navy), species (teal), and strain (red) resolution across 49 matched stool-diet

(strain *sp900542495*; $p = 0.606$; raw $P = 5.0 \times 10^{-6}$; Benjamini-Hochberg (BH)-adjusted $P = .037$). Importantly, dietary records show that alcohol intake in this cohort was largely derived from condiments within the study meals (eg, wine vinegars, vanilla), not from beverage consumption, as no alcoholic beverages were reported.

Microbial pathways linked with SCFA production are differentially abundant in the high consumers of plant-based diets

Next, we identified metabolic pathways that were differentially present in the high and low consumers of delivered meals at both the neutropenic nadir period and at the EOS (Figure 4A-B). Of note, the most significantly increased pathway in the high consumers at both time points was the aminobutanoate degradation pathway (PWY5022), which directly results in SCFA production.

We next measured SCFA in stool samples from baseline and at the EOS. We observed an increase in SCFA within the patient cohort, with caproic and isovaleric acid reaching statistical significance ($P = .023$ and $.05$, respectively), and valeric, isobutyric, and 2-methylbutyric acid reaching near-significant values ($P = .08$, $.06$, and $.07$; Figure 4C). Expectedly, given the small patient numbers, we did not observe differences in the measured metabolites when we compared the low and high consumption groups.

Patient-reported outcome surveys reveal that role functioning improves throughout the auto-HCT process, whereas other aspects of QoL remain unchanged or worsen

Patient-reported outcomes were also measured using the myeloma-specific MY20 and general cancer-related C30 tools and confirmed that some aspects of QoL improved throughout the study period. The MY20 is a 20-item, myeloma-specific QoL questionnaire used alongside the C30 to assess disease symptoms, treatment side effects, future perspective, and body image in patients with multiple myeloma. Each domain is scored from 0 to 100, and higher scores on symptom and side effect scales indicate worse symptom burden, whereas higher scores on functioning-oriented scales (eg, future perspectives, role functioning) indicate better QoL. Analysis of the MY20 revealed an increase in disease symptom scores and an improvement in scores related to side effects of treatment and body image throughout the study period. Scores for future perspectives were unchanged. The C30 survey demonstrated that scores for global health status/QoL declined between the baseline assessment and the EOS, but there was a modest improvement in physical and social functioning, and a substantial improvement in role functioning. The role functioning metric captures limitations in work,

household tasks, and other usual activities due to physical or emotional barriers. Specific scores are summarized in supplemental Table 4. Overall, the patient-reported outcomes provide valuable insights into the patient experience of auto-HCT, and we will build on these findings in future studies.

Discussion

This pilot study demonstrates that a home-delivered, plant-based meal intervention during auto-HCT is feasible and generally well tolerated, though strategies to improve adherence will be important for future trials. Higher meal consumption was associated with improved calorie and fiber intake, favorable shifts in select commensal gut taxa, and enrichment of microbial pathways linked to SCFA production. Together, these findings support the potential of targeted, plant-based nutritional strategies to modulate the gut microbiome and metabolic outputs during auto-HCT, warranting confirmation in larger, controlled studies.

The intervention met its primary feasibility end point, demonstrating that patients undergoing auto-HCT can participate in a plant-based whole-foods meal program during intensive treatment. Although gastrointestinal symptoms did limit intake during the nadir period, most patients recovered, and oral intake increased with count recovery.

The low HEI scores highlight the importance of dietary intervention strategies in this patient population as well as the challenge in improving dietary quality and diversity from a low baseline. Our study population had an HEI score in the “very poor” range throughout the study period, and although this is in line with national US averages, it is likely that the pretransplant diet quality reported in our patients is influenced by additional factors unique to this patient population. Patients commencing the auto-HCT process have typically just completed induction-phase therapy and are experiencing residual gastrointestinal toxicities, and many must reside away from home to be treated at our specialized cancer center and thus have limited access to cooking facilities.

Although this study was small and microbial and metabolic studies were exploratory, we still observed several changes within the study participants both over time and as a function of meal consumption. Consumption of the delivered plant-based meals was associated with measurable shifts in gut microbiome composition, with the largest changes observed among participants with the highest intake of plant-based study meals. The analyses of paired diet:microbiome data, derived from specific sample-associated dietary records independent of the binary overall consumer classification, provide complementary evidence that dietary fiber shapes specific microbial populations during post-HCT recovery, particularly within the Lachnospiraceae family. From a functional perspective, EOS fecal samples demonstrated an increase in SCFA within the intestinal tract, which suggests dietary intake can

Figure 3 (continued) sample pairs, from 16 unique participants. Each dot represents 1 taxon; the x-axis shows the Spearman ρ correlation coefficient and the y-axis shows $-\log_{10}$ (unadjusted P value). Nonsignificant taxa ($P \geq .05$) are shown in gray and significant taxa ($P < .05$) are colored by taxonomic level and sized by mean relative abundance. The horizontal dotted line indicates $P = .05$. Labels are shown for deduplicated significant hits (top 20 at most resolved rank per lineage retained + the 6 most prominent genera are numbered). Taxa are named in supplemental Table 1. (C) Bubble heat-matrix of the top 30 nutrient $\times$ taxa associations across all tested families, ranked by raw P value. Rows are nutrients (faceted on the left by category) and columns are individual taxa (faceted on top by family). Each bubble shows 1 Spearman correlation between dietary intake and stool taxon abundance. Bubble fill encodes the signed correlation coefficient (ρ ; green = negative, red = positive); bubble size encodes $-\log_{10}$ (raw P). All features were resolved to strain level (taxa named in supplemental Table 2). Panel A was created in BioRender. Markey K. (2026) <https://BioRender.com/84frknd>.

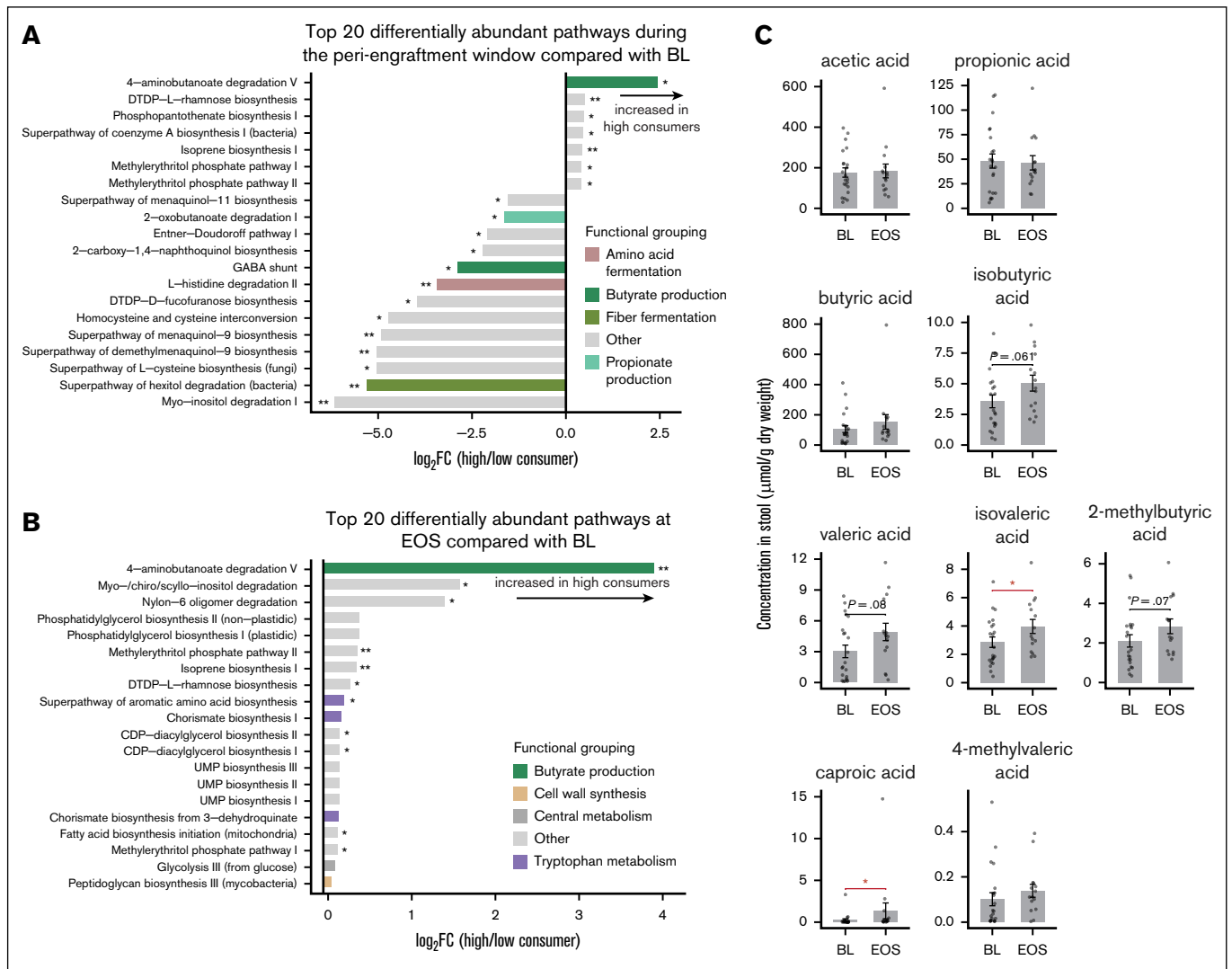


Figure 4. Stool SCFA are increased at the EOS period compared with BL samples. (A) Top 20 differentially abundant HUMAN2-annotated metabolic pathways during the neutropenic nadir window (high/low meal consumers). Bars extending to the right indicate pathways with higher abundance in high consumers and those extending toward the left indicate pathways with higher abundance in low consumers. Pathways were tested using Wilcoxon rank-sum test on copies-per-million abundance values, unadjusted P values ($*P < .05$, $**P < .01$). Sample sizes: high $n = 6$ to 7 , low $n = 5$ per time point. (B) Top 20 differentially abundant pathways at EOS ranked by $\log_2$ fold-change. The top-ranked pathway, 4-aminobutanoate (γ -aminobutyric acid) degradation V ($\log_2\text{FC} = 3.96$; $P = .007$). (B-C) Correcting across all ~ 200 HUMAN2 pathways at each time point, no individual pathway reached adjusted significance (all Benjamini-Hochberg adjusted $P = .995$). Unadjusted P values are as shown. (C) SCFA were measured in stool samples at BL and at EOS. All 22 patients gave a BL stool sample and 15 completed an EOS sample. Wilcoxon nonparametric tests are shown ($*P < .05$, other P values as annotated).

modulate microbial community function in the setting of post-HCT tissue damage. SCFAs are known to support epithelial barrier integrity and modulate immune function, providing a mechanistic rationale for targeting the microbiome with diet during auto-HCT.

There were several limitations to our study. Firstly, although we have assessed that the intervention is tolerable overall, future studies will require strategies to improve intake during the neutropenic nadir, when gastrointestinal symptoms are at their maximum. These include increasing the availability of more familiar “comfort foods,” as well as soups, smoothies, and other nutritional supplements. Despite the predominantly outpatient auto-HCT approach at our center, there was a significant rate of hospital admission, during which time access to the delivered meals was

limited, thus we also need to develop a strategy to continue meal support during inpatient hospitalization for future trials. Given the dose-dependent findings, we hypothesize that improving meal intake will lead to even greater nutritional and microbiome benefits. Establishing a required “dose” of plant-based meals will be an important component of future studies, as we work toward developing evidence-based dietary guidelines for patients.

We also acknowledge that this is a small, single-arm trial, and thus results are not definitive. The exploratory analyses, linking overall study meal consumption status with microbial changes and the paired diet:stool analyses, align well and generate valuable new hypotheses regarding the role of dietary nutrients in both supporting specific beneficial commensals and suppressing

pathobionts. Remarkably, despite thousands of tests and a high burden of correction for multiple comparisons, 1 correlation was robust to FDR adjustment. The robust correlation between alcohol in condiments and a strain of *Roseburia* perhaps suggests that home-style meals containing multiple condiments and sauces are an important dietary lever for maintenance of commensal Lachnospiraceae family members. We also acknowledge that the correlated structure of dietary nutrients (eg, fiber covaries with total energy intake) makes it difficult to attribute associations to individual nutrients without controlled feeding experiments. Additionally, a number of the findings at strain rank highlighted associations between dietary intake and strains of uncertain functional significance, making interpretation of some results challenging. Finally, in a single-arm study of this size, it is not possible to develop recommendations for patients based on these data.

Overall, we conclude that the provision of plant-based whole-food meals can alter the gut microbiome in a dose-responsive fashion and is associated with increased SCFA production, supporting larger efficacy-focused trials. In future studies, we propose incorporating behavioral and implementation science to optimize adherence and to test whether functional microbiome changes mediate improved transplant and myeloma outcomes.

Acknowledgments

The authors thank Tara Ohrazda, Farah Sahoo, and Chloe Te for their contributions to the start-up phase of the study.

This study was funded by a Hartwell Innovation Fund Swim Across America Pilot Award (K.A.M.). The authors acknowledge the support of the Cancer Center Support Grant to the Fred Hutchinson Cancer Center (P30 CA015704-50; NCI). A.C.Y. acknowledges funding from the American Society of Hematology Scholar Award and the National Institutes of Health K08 HL167161 (NHLBI).

The study did not receive sponsorship or funding from the meal delivery company Thistle; meals were purchased using independent study funds.

Authorship

Contribution: K.U. was the lead dietitian on the study, was involved in recruitment, monitoring, data collection and analysis, and contributed to writing the manuscript; T.E. was the lead study coordinator and contributed to recruitment, data collection, and analysis; M.R. cared for patients throughout the study; A.E.W. analyzed

microbiome data; H. Purcell analyzed metabolomics data; S.R.C. and M.M.-M. processed clinical samples; H. Persinger ran the biobanking protocol on which nonstudy auto-HCT control participants and collected stool samples; K.J. and A.B. contributed to study design and the 24-hour recall assessments; K.M. and M. Byelykh contributed to study design and edited the manuscript; M.A.B., A.C.Y., N.A., S.M., N.W., and L.A.H. cared for patients, aided in the screening process, and supported protocol development; D.N.F. supported protocol development and edited the manuscript; M. Bleakley supported the biobanking protocol and edited the manuscript; J.M.S. and M.L.N. contributed to protocol design, data analysis strategies, and edited the manuscript; D.R. supervised the metabolomics analysis and guided experimental design; J.U.P. contributed to design of analysis strategies and edited the manuscript; J.M. served as the trial biostatistician; G.R.H. supported study development, assisted in screening and recruitment, cared for patients, and edited the manuscript; S.J.L. supported the study design, helped developed the protocol, led the regulatory and coordinator team, and edited the manuscript; and K.A.M. designed the study, wrote the protocol, secured funding, consented patients, analyzed data, and wrote the manuscript.

Conflict-of-interest disclosure: M.R. reports consulting role for Johnson and Johnson. N.W. is on the advisory board for Mustang Bio and Bristol Myers Squibb. J.U.P. received research funding, intellectual property fees, and travel reimbursement from Seres Therapeutics; received consulting fees from Da Volterra, CSL Behring, Crestone Inc, MaaT Pharma, and Canaccord Genuity Inc and RA Capital; serves on an advisory board of and holds equity in Postbiotics Plus and Prodigy Biosciences; and has filed intellectual property applications related to the microbiome (reference numbers 62/843,849; /977,908; and 15/756,845). Memorial Sloan Kettering Cancer Center has financial interests relative to Seres Therapeutics. K.A.M. holds equity in Postbiotics Plus; and has previously consulted for Incyte and Crestone. The remaining authors declare no competing financial interests.

ORCID profiles: S.R.C., [0000-0003-3945-2496](#); H. Persinger, [0000-0002-4398-8616](#); A.C.Y., [0000-0003-3455-7360](#); N.W., [0000-0002-1010-4589](#); M. Bleakley, [0000-0002-7018-8702](#); J.U.P., [0000-0002-4029-7625](#); J.M.S., [0000-0003-4865-7813](#); J.M., [0000-0001-6294-227X](#); S.J.L., [0000-0003-2600-6390](#); K.A.M., [0000-0002-7064-2362](#).

Correspondence: Kate A. Markey, Fred Hutchinson Cancer Center, 1100 Fairview Ave N – mailstop S3-204, Seattle, WA 98109; email: kmarkey@fredhutch.org.

References

- Shah UA, Parikh R, Castro F, Bellone M, Lesokhin AM. Dietary and microbiome evidence in multiple myeloma and other plasma cell disorders. *Leukemia*. 2023;37(5):964-980.
- Khan N, Lindner S, Gomes ALC, et al. Fecal microbiota diversity disruption and clinical outcomes after auto-HCT: a multicenter observational study. *Blood*. 2021;137(11):1527-1537.
- Montassier E, Al-Ghalith GA, Ward T, et al. Pretreatment gut microbiome predicts chemotherapy-related bloodstream infection. *Genome Med*. 2016; 8(1):49.
- Kusakabe S, Fukushima K, Maeda T, et al. Pre- and post-serial metagenomic analysis of gut microbiota as a prognostic factor in patients undergoing haematopoietic stem cell transplantation. *Br J Haematol*. 2020;188(3):438-449.

5. El Jurdi N, Filali-Mouhim A, Salem I, et al. Gastrointestinal microbiome and mycobiome changes during autologous transplantation for multiple myeloma: results of a prospective pilot study. *Biol Blood Marrow Transpl.* 2019;25(8):1511-1519.
6. Dunneram Y, Lee JY, Watling CZ, et al. Vegetarian diets and cancer risk: pooled analysis of 1.8 million women and men in nine prospective studies on three continents. *Br J Cancer.* 2026;134(8):1218-1229.
7. Shah UA, Cogrossi LL, Garces JJ, et al. A high-fiber plant-based diet in myeloma precursor disorders - results from the NUTRIVENTION clinical trial and preclinical Vk*MYC model. *Cancer Discov.* 2025;16(4):697-720.
8. Hu EA, Steffen LM, Coresh J, Appel LJ, Rebholz CM. Adherence to the Healthy Eating Index-2015 and other dietary patterns may reduce risk of cardiovascular disease, cardiovascular mortality, and all-cause mortality. *J Nutr.* 2020;150(2):312-321.
9. Lee DH, Fung TT, Tabung FK, et al. Prediagnosis dietary pattern and survival in patients with multiple myeloma. *Int J Cancer.* 2020;147(7):1823-1830.
10. Arends J, Baracos V, Bertz H, et al. ESPEN expert group recommendations for action against cancer-related malnutrition. *Clin Nutr.* 2017;36(5):1187-1196.
11. Aaronson NK, Ahmedzai S, Bergman B, et al. The European Organization for Research and Treatment of Cancer QLQ-C30: a quality-of-life instrument for use in international clinical trials in oncology. *J Natl Cancer Inst.* 1993;85(5):365-376.
12. Cocks K, Cohen D, Wisloff F, et al. An international field study of the reliability and validity of a disease-specific questionnaire module (the QLQ-MY20) in assessing the quality of life of patients with multiple myeloma. *Eur J Cancer.* 2007;43(11):1670-1678.
13. Liese AD, Krebs-Smith SM, Subar AF, et al. The Dietary Patterns Methods Project: synthesis of findings across cohorts and relevance to dietary guidance. *J Nutr.* 2015;145(3):393-402.
14. Reedy J, Krebs-Smith SM, Miller PE, et al. Higher diet quality is associated with decreased risk of all-cause, cardiovascular disease, and cancer mortality among older adults. *J Nutr.* 2014;144(6):881-889.
15. Shams-White MM, Pannucci TE, Lerman JL, et al. Healthy Eating Index-2020: review and update process to reflect the dietary guidelines for Americans, 2020-2025. *J Acad Nutr Diet.* 2023;123(9):1280-1288.
16. Fraipont V, Preiser JC. Energy estimation and measurement in critically ill patients. *JPEN J Parenter Enteral Nutr.* 2013;37(6):705-713.
17. Cotillard A, Cartier-Meheust A, Litwin NS, et al. A posteriori dietary patterns better explain variations of the gut microbiome than individual markers in the American Gut Project. *Am J Clin Nutr.* 2022;115(2):432-443.
18. Johnson AJ, Zheng JJ, Kang JW, Saboe A, Knights D, Zivkovic AM. A guide to diet-microbiome study design. *Front Nutr.* 2020;7:79.